ΕΠ1445 - Εργασία 3 · Μέρος Α

$I = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$, $N = M = 2$

$$\tilde{I}_{real}(u,v) = \sum_{i=0}^{1} \sum_{j=0}^{1} I(i,j) \cdot \cos\left(2\pi \cdot \left(\frac{ui}{2} + \frac{vj}{2}\right)\right)$$

$\sin(k \cdot \pi) = 0 \ \forall k$

$$\tilde{I}_{im}(u,v) = \sum_{i=0}^{1} \sum_{j=0}^{1} I(i,j) \cdot \sin\left(2\pi \cdot \left(\frac{ui}{2} + \frac{vj}{2}\right)\right) = \sum_{i=0}^{1} \sum_{j=0}^{1} I(i,j) \cdot \sin(\pi(i+j))$$

$$\boxed{= 0 \ \forall u,v}$$

$$\left|\tilde{I}(u,v)\right| = \sqrt{\tilde{I}_{real}(u,v)^2 + \tilde{I}_{im}(u,v)}$$

$$\angle \tilde{I}(u,v) = \tan^{-1}\left(\frac{\tilde{I}_{im}(u,v)}{\tilde{I}_{real}(u,v)}\right)$$

<u>Βήμα 1:</u> Υπολογισμός $\tilde{I}_{real} = \begin{bmatrix} 4 & 2 \\ 2 & 0 \end{bmatrix}$

- $$\tilde{I}_{real}(0,0) = \sum_{i=0}^{1} \sum_{j=0}^{1} I(i,j) \cdot \cos(2\pi \cdot 0)$$

$$= I(0,0) \cdot \cos 0 + I(0,1) \cdot \cos 0 + I(1,0) \cdot \cos 0 + I(1,1) \cdot \cos 0$$

$$= 2 + 1 + 1 + 0$$

$$\boxed{= 4}$$

- $$\tilde{I}_{real}(0,1) = \sum_{i=0}^{1} \sum_{j=0}^{1} I(i,j) \cdot \cos\left(2\pi \cdot \frac{j}{2}\right)$$

$$= I(0,0) \cdot \cos 0 + I(0,1) \cdot \cos(\pi) + I(1,0) \cdot \cos 0 + I(1,1) \cdot \cos(\pi)$$

$$= 2 - \cancel{1} + \cancel{1} - 0$$

$$\boxed{= 2}$$

- $$\tilde{I}_{real}(1,0) = \sum_{i=0}^{1} \sum_{j=0}^{1} I(i,j) \cdot \cos\left(2\pi \cdot \frac{i}{2}\right)$$

$$= I(0,0) \cdot \cos 0 + I(0,1) \cdot \cos 0 + I(1,0) \cos \pi + I(1,1) \cos \pi$$

$$= 2 + \cancel{1} - \cancel{1} - 0$$

$$\boxed{3}$$

- $$\tilde{I}_{real}(1,1) = \sum_{i=0}^{1} \sum_{j=0}^{1} I(i,j) \cdot \cos\left(2\pi\left(\frac{i+j}{2}\right)\right)$$

$$= I(0,0) \cdot \cos 0 + I(0,1) \cdot \cos \pi + I(1,0) \cdot \cos \pi + I(1,1) \cdot \cos 2\pi$$

$$= 2 - 1 - 1 + 0$$

$$\boxed{0}$$

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Βήμα 2: Υπολογισμός $\tilde{I}_{im} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

- $\tilde{I}_{im}(0,0) = \sum_{i=0}^{1} \sum_{j=0}^{1} I(i,j) \cdot \sin(2\pi \cdot 0)$

  $= 0$ (αφού $\sin 0 = 0$)

- $\tilde{I}_{im}(0,1) = \sum_{i=0}^{1} \sum_{j=0}^{1} I(i,j) \cdot \sin(2\pi \cdot j)$

  $= \cancel{I(0,0) \cdot \sin(0)}^{0} + \cancel{I(0,1) \sin(\pi)}^{0} + \cancel{I(1,0) \sin(0)}^{0} + \cancel{I(1,1) \sin(\pi)}^{0}$

  $= 0 + 0 + 0 + 0$

  $\boxed{=0}$

- $\tilde{I}(1,0) = \sum_{i=0}^{1} \sum_{j=0}^{1} I(i,j) \cdot \sin(2\pi \cdot i)$

  $= \cancel{I(0,0) \cdot \sin 0}^{0} + \cancel{I(0,1) \cdot \sin 0}^{0} + \cancel{I(1,0) \cdot \sin \pi}^{0} + \cancel{I(1,1) \cdot \sin \pi}^{0}$

  $= 0 + 0 + 0 + 0$

  $\boxed{=0}$

- $\tilde{I}(1,1) = \sum_{i=0}^{1} \sum_{j=0}^{1} I(i,j) \cdot \sin\left(2\pi \left(\frac{i+j}{2}\right)\right)$

  $= \cancel{I(0,0) \cdot \sin 0}^{0} + \cancel{I(0,1) \cdot \sin \pi}^{0} + \cancel{I(1,0) \cdot \sin \pi}^{0} + \cancel{I(1,1) \sin 2\pi}^{0}$

  $= 0 + 0 + 0 + 0$

  $\boxed{=0}$

Βήμα 3: Υπολογισμός $|\tilde{I}| = \begin{bmatrix} 4 & 2 \\ 2 & 0 \end{bmatrix}$

- $|\tilde{I}(0,0)| = \sqrt{\tilde{I}_{real}(0,0)^2 + \tilde{I}_{im}(0,0)^2} = \sqrt{16 + 0} = 4$
- $|\tilde{I}(0,1)| = \sqrt{\tilde{I}_{real}(0,1)^2 + \tilde{I}_{im}(0,1)^2} = \sqrt{4 + 0} = 2$
- $|\tilde{I}(1,0)| = \sqrt{\tilde{I}_{real}(1,0)^2 + \tilde{I}_{im}(1,0)^2} = \sqrt{4 + 0} = 2$
- $|\tilde{I}(1,1)| = \sqrt{\tilde{I}_{real}(1,1)^2 + \tilde{I}_{im}(1,1)^2} = \sqrt{0 + 0} = 0$

Βήμα 4: Υπολογισμός $\angle \tilde{I} = \begin{bmatrix} 0 & 0 \\ 0 & \end{bmatrix}$

- $\angle \tilde{I}(0,0) = \tan^{-1}(\tilde{I}_{im}(0,0) / \tilde{I}_{real}(0,0)) = \tan^{-1}(0/4) = 0$
- $\angle \tilde{I}(0,1) = \tan^{-1}(\tilde{I}_{im}(0,1) / \tilde{I}_{real}(0,1)) = \tan^{-1}(0/2) = 0$
- $\angle \tilde{I}(1,0) = \tan^{-1}(\tilde{I}_{im}(1,0) / \tilde{I}_{real}(1,0)) = \tan^{-1}(0/2) = 0$
- $\angle \tilde{I}(1,1) = \tan^{-1}(\tilde{I}_{im}(1,1) / \tilde{I}_{real}(1,1)) = \tan^{-1}(0/0) =$ Μηδέν (%)

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